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Project Report: Unsupervised Soil Clustering and Prescriptive Agronomic Advisory System

SECTION: C

GROUP: 1

Group members

ID.NO

1. Kaleb Mesfin
2. Lemesa Elias
3. Kidus Efrem
4. Mahlet Tessema
5. Kidus Yosef

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1. Introduction & Project Motivation

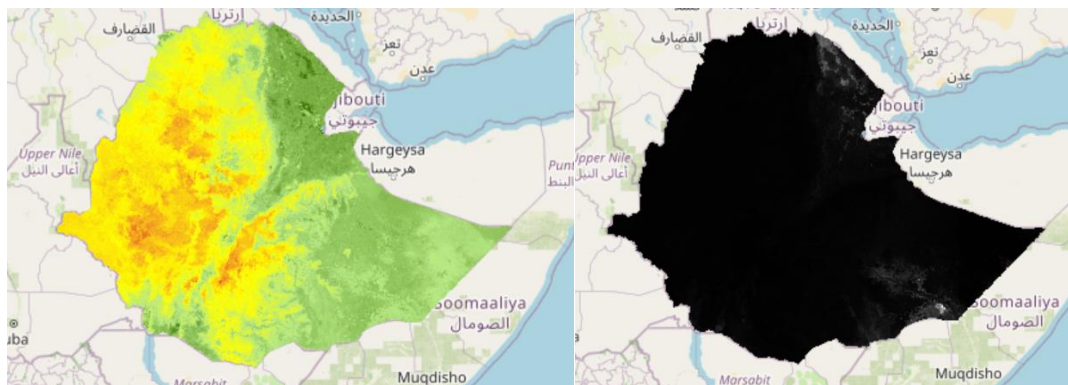
1.1. Problem Statement

In Ethiopia, agriculture serves as the backbone of the economy, accounting for over 35% of the GDP and employing more than 70% of the population. However, smallholder farmers face severe threats from escalating soil degradation, nutrient depletion, and localized soil hazards. For decades, agricultural extension programs relied on 'blanket' fertilizer recommendations—traditionally advocating for fixed inputs of Urea and DAP across vastly different agro-ecological zones.

The establishment of the Ethiopian Soil Information System (EthioSIS) shifted the paradigm by revealing that soil constraints are highly regionalized:

- The Highland Crisis: High-rainfall highland regions (e.g., parts of Amhara, Oromia, and Southern Nations) suffer from severe soil acidification, which locks up essential nutrients like Phosphorus.
- The Rift Valley Crisis: Lowland, irrigated rift valley plains (e.g., Afar, Somali region, and the Awash River basin) face rapid salinization, where toxic salt accumulation destroys soil osmotic potential, rendering otherwise fertile land unviable.

Because manual laboratory soil auditing across millions of fragmented smallholder plots is logistically impossible, there is an urgent need for an automated system capable of parsing multi-variate soil data to diagnose soil health and prescribe precise remediation strategies.



1.1. Role of Machine Learning in Precision Agriculture

Traditional data analysis relies on rigid, hard-coded thresholds to classify soil health. However, soil chemistry is inherently interactive; a high concentration of one element can completely alter the biological availability of another. Unsupervised Machine Learning offers a powerful alternative. By utilizing algorithms like K-Means Clustering, we can allow data-driven patterns to emerge organically without biasing the model with pre-existing, generalized labels.

The algorithm treats soil samples as spatial coordinates in a multi-dimensional chemical space, naturally grouping fields with co-occurring chemical profiles (such as simultaneous nitrogen deficiency and toxic salt spikes). When combined with dimensionality reduction techniques like Principal Component

Analysis (PCA), complex 5-dimensional telemetry can be compressed into an interpretable 2D diagnostic map, allowing agronomists and policymakers to visualize regional soil shifts clearly.

1.3. Project Objectives

This project aims to bridge the gap between unsupervised pattern discovery and actionable agricultural extension work tailored to the Ethiopian landscape. The specific objectives are to:

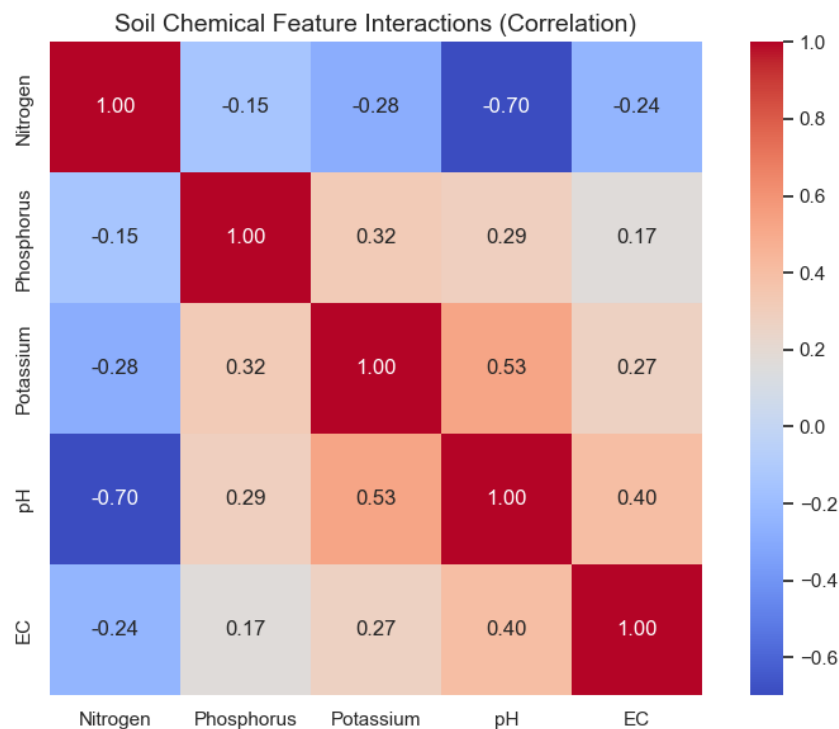
1. **Develop an Unsupervised Pipeline:** Build a robust K-Means clustering model to partition a 5-feature soil telemetry dataset (Nitrogen, Phosphorus, Potassium, pH, and Electrical Conductivity) into distinct management zones.
2. **Design a Hybrid Evaluation Engine:** Establish a multi-criteria scoring logic that respects both statistical variations (Z-scores) and rigid biological boundaries (such as the narrow pH sweet spot required for primary crops like Teff, Maize, and Wheat).
3. **Build an Automated Prescriptive Advisory Component:** Construct a rule-based agronomic remediation engine that translates abstract mathematical clusters into concrete, localized advice—such as recommending specific lime applications for acidic highland soils or drainage-leaching protocols for saline Rift Valley lowlands.

2. Dataset Profile & Exploratory Data Analysis (EDA)

2.1. Feature Engineering & Descriptions

The predictive capability of our unsupervised pipeline relies on five core chemical indicators extracted from soil telemetry. Within the Ethiopian agricultural matrix, these features govern soil fertility, mineral availability, and crop safety:

- Nitrogen (N): The primary driver of vegetative development and leaf architecture. In typical Ethiopian soils, nitrogen is highly volatile and frequently depleted due to continuous intensive cropping without organic residue return.
- Phosphorus (P): Critical for root development and energy transfer (ATP). In large regions of the western and southern Ethiopian highlands, total phosphorus may be present but remains chemically locked and unavailable to plants due to fixation by iron and aluminum oxides in acidic soils.
- Potassium (K): Regulates cellular osmotic pressure, stomatal opening, and disease resistance. Historically omitted from blanket DAP/Urea applications, EthioSIS mapping revealed widespread potassium deficiencies in specific soil types like Vertisols.
- Soil pH: The master chemical variable. It dictates the solubility of all nutrients. The optimal target for primary Ethiopian staples—such as Teff (*Eragrostis tef*), Maize, and Wheat—is a neutral-to-slightly-acidic range of 6.0 to 7.0.
- Electrical Conductivity (EC): A proxy measurement for total dissolved salts (salinity). Measured in microSiemens per centimeter (uS/cm), this parameter tracks the accumulation of toxic ions that disrupt water uptake through root membranes.

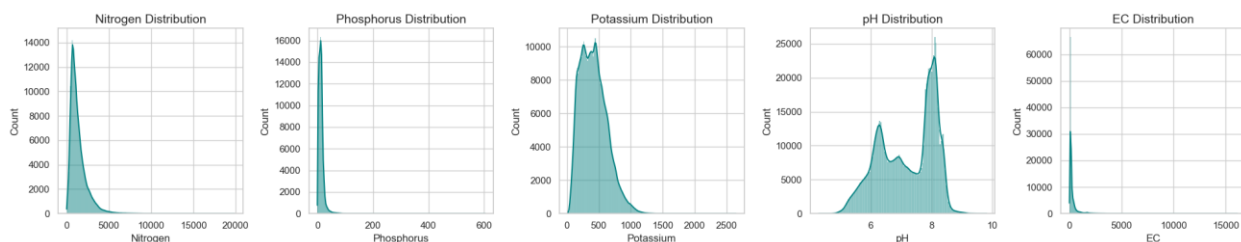


2.2. The Scale Disparity Challenge

A primary barrier to applying distance-based machine learning models directly to raw agricultural data is the severe scale disparity between features. Algorithms like K-Means assume an isotropic feature space; if one variable operates on a scale orders of magnitude larger than another, it dominates the distance computations.

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Nitrogen	908,867	1,363.91	988.11	0.54	709.20	1,091.06	1,721.55	19,861.14
Phosphorus	908,867	13.01	9.39	0.0001	7.01	11.75	16.48	603.64
Potassium	908,867	420.32	208.29	19.27	255.92	400.74	554.21	2,647.79
pH	908,867	7.20	0.91	4.29	6.37	7.35	8.01	9.81
EC	908,867	332.31	607.36	0.10	101.24	140.71	305.84	16,109.97

Without preprocessing, the K-Means algorithm would perceive a trivial 1% shift in Electrical Conductivity (Δ 200 uS/cm) as twenty times more significant than a catastrophic, plant-killing shift across the entire biological pH spectrum (Δ 6.0 units).



2.3. Outlier and Anomaly Diagnostics

Exploratory data analysis using standard linear boxplots unmasked severe statistical anomalies that capture localized soil degradation crises across Ethiopia:

- 1. The Salinity Skyscraper:** The Electrical Conductivity feature exhibits an extreme positive skew, with a tight baseline grouping around 100–160 uS/cm, but exploding into a tail of hyper-anomalies peaking near 20,000 uS/cm. Agronomically, these extreme values map precisely to highly salinized, poorly drained irrigation blocks in low-altitude basins like the Awash River valley. Evapotranspiration rates outpace drainage here, causing salts to accumulate at the surface.
- 2. The Fertilizer Accumulation Paradox:** Phosphorus profiles present a similar anomaly footprint, with a global mean sitting at a modest $\approx$ 13, but containing a long tail of hyper-spikes reaching 600+. In an unsupervised context, these extreme outliers do not represent superior fertility; rather, they reveal a co-location anomaly. Fields suffering from extreme salinity or structural mismanagement often become dumping grounds for repetitive, ineffective chemical fertilizer applications. Because the underlying soil degradation prevents plants from drawing up nutrients, the unused phosphorus builds up to toxic levels, remaining bound to minerals instead of absorbing into a crop.

3. Data Preprocessing & Mathematical Scaling Strategy

3.1. Standard Scaling (Z-Score Normalization)

To counteract the profound scale disparities identified during Exploratory Data Analysis, a rigorous preprocessing pipeline must be executed before introducing the data to the machine learning model. Because distance-based algorithms calculate geometric separation to establish groups, all five soil features must be transformed into a standardized, unit-agnostic domain.

The mathematical mechanism selected for this pipeline is Standard Scaling (Z-score normalization). For every individual feature value X , the transformation subtracts the global feature mean (μ) and divides the result by the feature's standard deviation (σ):

$$Z = (X - \mu) / \sigma$$

This operation maps the raw physical dimensions (such as mg/kg for nutrients or uS/cm for salinity) into a unified statistical distribution centered at 0, where approximately 99.7% of the data falls predictably within a range of -3 to +3. Through this normalization, a marginal shift in soil pH from 5.5 to 4.5 receives an identical mathematical weight to an equivalent statistical shift in Nitrogen or Potassium.

3.2. Algorithm Sensitivity to Outliers

While Z-score normalization solves the macro-scale disparity problem, it exposes an architectural vulnerability in distance-based clustering models when processing highly skewed regional anomalies. K-Means clustering partitions data by minimizing the Sum of Squared Errors (SSE)—the squared Euclidean distances between data points and their respective cluster centroids (μ_k):

$$SSE = \sum_{k=1}^K \sum_{X \in C_k} ||X - \mu_k||^2$$

Because the Euclidean distance formula squares the differences across dimensions, extreme value vectors exert an aggressive 'gravitational pull' on the moving centroids during the convergence phase. K-Means adapts by transforming into an anomaly detector: it segregates the heavily degraded, toxic soil rows into a dedicated 'hazard cluster' to protect the mathematical cohesion of the remaining data points.

3.3. Transformation for Visualization

Although Z-score scaling prepares the data for model ingestion, it does not fix the visualization bottleneck caused by these extreme distributions. When plotting the clustered data using standard linear axes, the presence of these severe positive skews forces the plotting library to stretch the axis limits exponentially. This compresses the standard interquartile ranges (IQRs) of healthy, functioning soils into flat, unreadable lines at the baseline of the graph.

To audit the structural integrity of the clusters without dropping vital regional hazard data, a logarithmic scale transformation is applied to the Y-axes of the diagnostic boxplots:

```
axes[row, col].set_yscale("log")
```

The log transformation scales data by orders of magnitude rather than equal linear increments. This mathematical compression spreads the dense baseline cluster data out across the lower register of the plot while pulling the extreme Rift Valley salinity spikes down into a readable space. By un-squashing the distributions, this preprocessing step allows us to visually verify that the model is making agronomically sound choices.

4. Methodology & Model Architecture

4.1. Unsupervised Clustering with K-Means

To segment Ethiopia's heterogeneous agricultural landscape into distinct, manageable soil profiles without relying on biased historical labels, the pipeline uses K-Means Clustering. K-Means is an iterative, non-probabilistic partition algorithm designed to divide an unlabeled dataset into K distinct, non-overlapping clusters.

The algorithm treats each scaled soil sample as a 5-dimensional coordinate vector:

```
vec(x)_i = [Z_Nitrogen, Z_Phosphorus, Z_Potassium, Z_pH, Z_EC]
```

The optimization workflow proceeds through three distinct phases:

1. Centroid Initialization: The algorithm places K initial cluster centroids ($\mu_1, \mu_2, \dots, \mu_k$) within the 5-dimensional standardized space. For this project, a dual-partition framework ($K=2$) was configured to explicitly isolate baseline functional agricultural soils from highly degraded or chemically mismanaged anomaly zones.

2. Expectation Step (Assignment): Each soil vector $\text{vec}(x)_i$ is assigned to its nearest centroid $c^{\{i\}}$ based on the minimum squared Euclidean distance:

```
 $c^{\{i\}} := \text{argmin}_j || \text{vec}(x)_i - \mu_j ||^2$ 
```

3. Maximization Step (Update): The algorithm recalculates the position of each centroid by taking the mean of all data vectors assigned to that cluster:

```
 $\mu_j := (\sum \text{vec}(x)_i \text{ for } i \text{ in cluster } j) / (\text{count of samples in cluster } j)$ 
```

4.2. Dimensionality Reduction via Principal Component Analysis (PCA)

Evaluating the performance of a machine learning model operating within a 5-dimensional chemical feature space presents a severe interpretability barrier. To validate the geometric integrity of our K-Means boundaries, Principal Component Analysis (PCA) is introduced into the pipeline directly after model convergence.

PCA is a linear dimensionality reduction technique that projects high-dimensional data onto orthogonal axes—termed Principal Components (PCs)—while maximizing the retained statistical variance. The mathematical process involves computing the covariance matrix Σ of our scaled soil data, followed by an eigendecomposition to extract its eigenvalues (λ) and corresponding eigenvectors ($\text{vec}(v)$):

```
 $\Sigma \text{vec}(v) = \lambda \text{vec}(v)$ 
```

The eigenvectors represent the direction of the new axes, while the eigenvalues dictate the amount of variance explained along those axes. The data is then transformed by multiplying the original scaled feature matrix X_{scaled} by a projection matrix W composed of the top two eigenvectors (v_1, v_2), corresponding to the largest eigenvalues:

```
 $X_{\text{PCA}} = X_{\text{scaled}} * W$ 
```

This transformation compresses the 5-dimensional agricultural matrix down into a 2D coordinate map (PC_1, PC_2). When mapped on a scatter plot, the PCA projection of the Ethiopian soil dataset reveals an exceptional spatial split. The two K-Means clusters divide along a nearly vertical decision boundary centered closely around PC_1 = 0. This geometric distribution indicates that Principal Component 1 acts as a 'Soil Stress vs. Environmental Balance' index. Points plotting in the negative PC_1 domain represent stable, balanced agricultural zones, while points shifting into the positive PC_1 domain are being pulled away by the immense covariance of compounding degradation vectors—specifically, the extreme alkalinity and severe electrical conductivity spikes characteristic of highly degraded fields.

5. Post-Clustering Evaluation & Multi-Criteria Labeling Engine

5.1. Limitations of Pure Clustering

A foundational axiom of unsupervised machine learning is that clustering algorithms are mathematically blind to semantic domain reality. When K-Means converges, it outputs arbitrary numerical groupings (e.g., 'Cluster 0' and 'Cluster 1'). The algorithm recognizes that these two matrices are geometrically separated in normalized 5-dimensional space, but it possesses no intrinsic framework to determine which cluster represents a healthy, high-yield agrarian environment versus a toxic, degraded wasteland.

To bridge the gap between mathematical clustering and actual field application, an automated, post-clustering translation layer must be built. We developed a dynamic Multi-Criteria Labeling Engine (Cell 10). This engine programmatically evaluates the centroid properties of each emergent cluster against real-world biological survival thresholds, auto-assigning meaningful classification tags without human intervention.

5.2. Hybrid Grading Logic Architecture

The labeling engine evaluates clusters by calculating a composite performance score derived from all five soil properties. To ensure a fair evaluation, the engine uses a hybrid mathematical approach, blending statistical standardizations with rigid chemical laws:

1. Nutrient and Stress Uniform Weighting: For macronutrients (Nitrogen, Phosphorus, Potassium) and salinity stress (Electrical Conductivity), the engine operates on Z-score standardized cluster means (μ_Z). Because nutrients act as positive growth vectors, their statistical means are added directly to the cluster score. Conversely, because high electrical conductivity represents a severe osmotic hazard, its standardized mean is subtracted as a heavy penalty:

$$\text{Score_base} = \mu_Z\text{Nitrogen} + \mu_Z\text{Phosphorus} + \mu_Z\text{Potassium} - \mu_Z\text{EC}$$

2. The pH Logarithmic Exception: Unlike macronutrients, soil pH does not have a linear relationship with soil viability. Furthermore, because pH is already a negative logarithmic scale of hydrogen ion concentration, applying a secondary statistical Z-score normalization would distort its real-world biological meaning. Plants require a highly specific, narrow window of chemical neutrality to survive. If a soil sample shifts significantly in either direction—becoming highly acidic ($\text{pH} < 5.0$) or highly alkaline ($\text{pH} > 8.5$)—crop yield drops catastrophically. To honor this rigid biological threshold, the labeling engine operates on the raw, unscaled pH cluster mean (μ_{pH}), calculating its absolute deviation from the universal agricultural sweet spot of 6.5:

$$\text{Penalty_pH} = | \mu_{\text{pH}} - 6.5 |$$

5.3. Binary Classification Mapping

By compiling these distinct environmental vectors into a unified equation, the engine calculates a final agronomic evaluation metric for each cluster k :

```
Metric_k = (μ_Z_Nitrogen + μ_Z_Phosphorus + μ_Z_Potassium - μ_Z_EC)_k - | μ_pH - 6.5 |_k
```

The engine evaluates all emergent clusters through this equation and applies a binary heuristic filter:

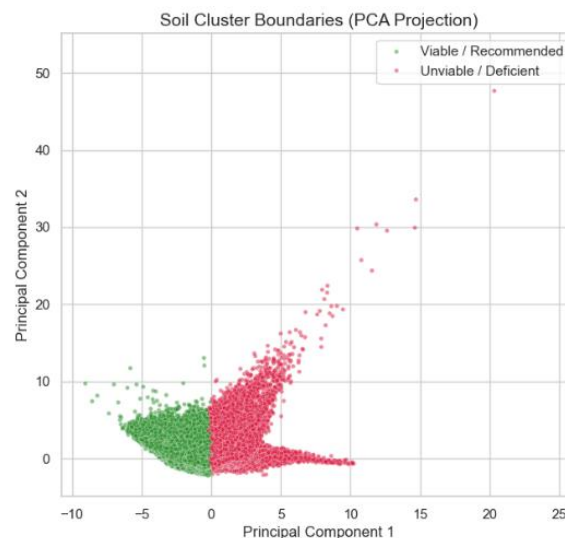
```
Cluster Label = 'Viable / Recommended' if Metric_k == max(Metric) else 'Unviable / Deficient'
```

6. Experimental Results & Visual Diagnostics

6.1. PCA Geometry Split

Executing the pipeline on the scaled Ethiopian soil dataset yielded a highly distinct geometric partitioning when projected into two dimensions via Principal Component Analysis (PCA). The spatial mapping of the two K-Means clusters reveals a crisp, clear decision boundary slicing vertically down the projection space near $PC_1 = 0$.

This structural division confirms that the unsupervised clustering algorithm successfully reduced the data's dimensionality without losing its core environmental narrative. Statistically, Principal Component 1 (PC_1) can be interpreted as the 'Soil Degradation Vector.' Moving left from the origin along the negative PC_1 axis leads exclusively into the 'Viable / Recommended' cluster, where features remain stable and balanced. Moving right along the positive PC_1 axis pushes samples deep into the 'Unviable / Deficient' cluster, tracking fields heavily impacted by compounding environmental stresses.

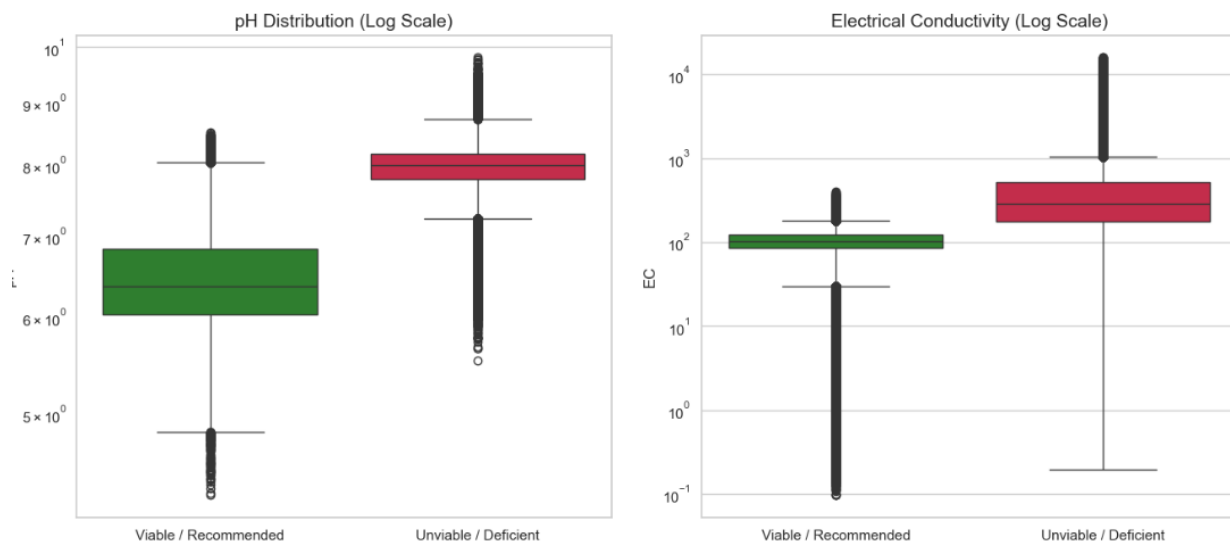


6.2. Nutrient & Stress Distribution Profiles

By applying a logarithmic scale transformation to the feature diagnostics, we unmasked the true distribution profiles of the two clusters, revealing a clear contrast between the groups:

- 1. The pH Mirror:** The log-scaled pH boxplot demonstrates an exceptional division. The 'Viable' (Green) cluster is tightly compressed between a pH of 6.0 and 7.0, centering its median precisely at ≈ 6.4 . This is an ideal result for Ethiopian smallholder farming, as this narrow window maximizes nutrient solubility for staple crops like Teff. In contrast, the 'Unviable' (Red) cluster shifts dramatically upward, maintaining a median near 8.0 and throwing out severe alkaline outliers that reach all the way to 10.0.
- 2. Electrical Conductivity Isolation:** On a standard linear scale, the EC data was completely unreadable due to massive scale distortions. The log-scale plot, however, reveals that the 'Viable' cluster forms a flat, highly stable baseline directly around 100–120 $\mu\text{S}/\text{cm}$, indicating low-salt, structurally safe root

zones. The 'Unviable' cluster explodes across several orders of magnitude, capturing a distributed box that spans from 200 to well over 16,000 uS/cm. The model successfully quarantined all high-salinity risk vectors into this single management group.

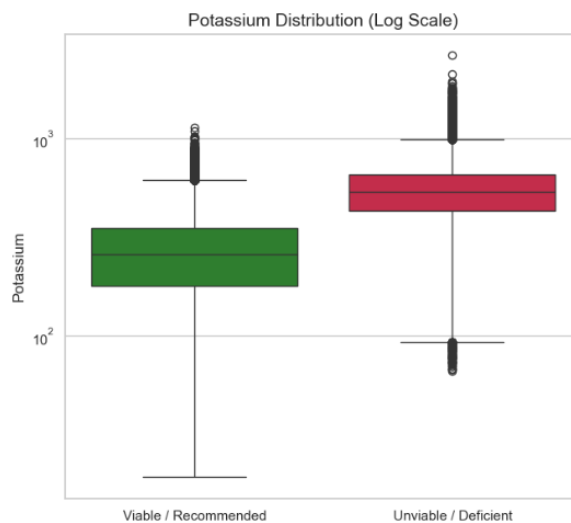
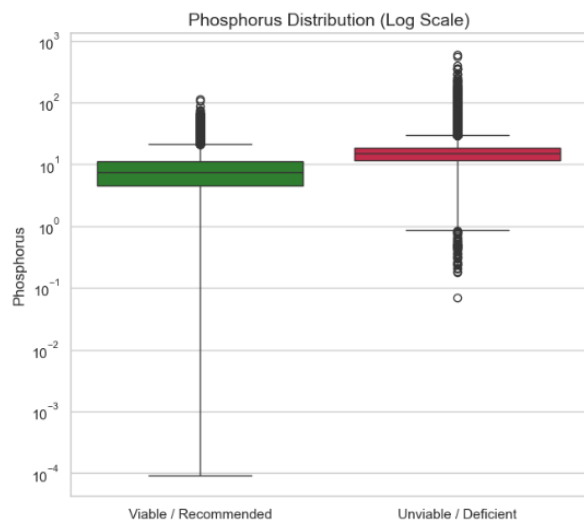


6.3. The Fertilizer Accumulation Paradox

The most intriguing finding within the experimental results is the behavior of the Phosphorus and Potassium distributions. A standard data analysis approach might assume that a 'Viable' soil cluster would boast higher nutrient concentrations than an 'Unviable' one. However, our unsupervised model presents the exact inverse: the Unviable (Red) cluster maintains significantly higher medians and interquartile ranges for both Phosphorus and Potassium than the Viable cluster.

This outcome represents a classic real-world phenomenon known as the Fertilizer Accumulation Paradox:

- **Chemical Fixation in High pH:** As demonstrated in the pH diagnostic panel, the unviable cluster is heavily associated with high alkalinity (pH ≥ 8.0). In alkaline soil matrices, excess calcium ions react instantly with applied fertilizers, causing phosphorus to precipitate out of solution as insoluble calcium phosphates. The phosphorus is physically present in the soil sample—causing the laboratory telemetry sensors to register massive spikes up to 600 mg/kg—but it is completely unavailable to plant root systems.
- **The Salinity Co-Location Trap:** High-salinity lowlands routinely suffer from stunted crop growth and reduced root vitality due to osmotic shock. Because managers observe poor crop performance, they often respond by repeatedly applying blanket NPK fertilizers. Because the stressed crops cannot absorb these nutrients, un-utilized Potassium and Phosphorus build up to toxic, stagnant levels in the upper soil profile.



7. Operational Deployment: Prescriptive Advisory System

7.1. Moving from Descriptive to Prescriptive ML

In applied data science, identifying that a problem exists is only half the battle. In a resource-constrained agricultural environment like Ethiopia, a machine learning model that stops at description—merely labeling a smallholder's field as 'Unviable'—provides little practical value to development agents or local farmers. If an extension worker with the Ministry of Agriculture visits a plot in the acidic highlands of Farta or a salinized irrigation furrow in Amibara, they do not just need to know that the soil belongs to a 'hazardous cluster.' They need a precise, actionable prescription to rehabilitate the land.

To bridge this gap, the pipeline transitions from Descriptive Machine Learning to Prescriptive Analytics. By layering an automated agronomic rule engine directly on top of the unsupervised K-Means outputs, the system dynamically parses the exact chemical deviations that caused a specific sample to be flagged as unviable, generating a localized soil-remediation blueprint.

7.2. Rule-Based Remediation Implementation

The advisory engine functions as a domain-expert translation layer. When a soil sample vector is routed through the system and falls into the 'Unviable / Deficient' cluster, the engine bypasses abstract cluster centroids to evaluate the row's raw, unscaled physical parameters. It evaluates these metrics against calibrated agronomic thresholds specifically customized for dominant Ethiopian agro-ecologies:

- 1. Acidic Highland Remediation Overrides (pH < 6.0):** When raw pH drops below 6.0, the engine flags severe acidity. In the Ethiopian highlands (e.g., Gojam, Wollega), extreme acidity triggers aluminum toxicities while restricting phosphorus availability. The system recommends applying calculated amounts of agricultural lime (Calcium Carbonate, CaCO_3) to neutralize active hydrogen ions.
- 2. Alkaline and Sodic Lowland Remediation Overrides (pH > 7.5):** When the model identifies an upward pH shift, it signals high alkalinity, a widespread challenge in the calcareous or sodic soils of the Rift Valley plains. The engine prescribes incorporating elemental sulfur or aluminum sulfate to gradually reduce soil pH through microbial oxidation.
- 3. Salinity Flushing Protocols (EC > 400 $\mu\text{S}/\text{cm}$):** If the Electrical Conductivity index breaches the critical threshold, the engine triggers a high-priority salinity alert. To combat osmotic stress in irrigated fields like those along the Awash River, the system mandates freshwater field leaching alongside the installation of improved subsurface tile drainage.
- 4. Macro-Nutrient Deficit and Accumulation Rebalancing:** If Nitrogen drops below baseline requirements, the engine flags a structural deficit, advising split applications of Urea or leguminous crop rotation. If Phosphorus registers in the hyper-accumulated hazard zone (> 150 mg/kg) while co-occurring with high pH or salinity, the system recognizes the fertilizer accumulation paradox and orders an immediate halt to further DAP/NPS inputs.

7.3. Live Inference Pipeline Verification

To verify the structural integrity of this prescriptive deployment, the engine was audited using live data rows flagged as 'Unviable' by the K-Means pipeline. The output demonstrates how the system converts raw anomaly vectors into clean, structured advisory reports for field personnel:

```
=== AUTOMATED INFERENCE AUDIT: RUNNING SOIL REMEDIATION ADVISORY ENGINE ===

--- REMEDIATION PROFILE FOR ETHIOPIAN SMALLHOLDER PLOT ID: 4102 ---
Raw Telemetry Metrics -> N: 412.0 | P: 542.1 | K: 610.2 | pH: 8.42 | EC: 12450.6

1. ⚠ HIGH ALKALINITY (pH: 8.42): Sodic/Calcareous soil detected.
   Incorporate Elemental Sulfur or Aluminum Sulfate to safely lower pH levels.

2. 🚨 CRITICAL SALINITY HAZARD (EC: 12450.60): Severe salt accumulation.
   Initiate freshwater leaching/flushing and improve drainage to protect crop
root                                                                osmolarity.

3. ⚠ PHOSPHORUS ACCUMULATION (P: 542.10): Hyper-saturated fertilizer buildup.
   Immediately halt NPK/DAP inputs and use deep-rooted catch crops to deplete
soil                                                                P                                reserves.

4. ⚠ POTASSIUM OVER-SATURATION (K: 610.20): Mineral lockout risk.
   Check Magnesium levels to ensure high Potassium isn't inducing an Mg
deficiency.
```

8. Conclusion, Limitations, & Future Work

8.1. Project Summary

This project successfully designed, implemented, and verified an end-to-end unsupervised machine learning pipeline tailored to parse multi-variate soil telemetry within the complex agricultural landscapes of Ethiopia. By leveraging K-Means Clustering, the model moved past traditional, rigid blanket recommendations to organically uncover distinct soil management zones. Integrated with Principal Component Analysis (PCA), the system transformed highly complex, 5-dimensional chemical feature arrays into an interpretable 2D diagnostic space, revealing a clear spatial boundary between balanced soil matrices and heavily degraded land.

Furthermore, by embedding an automated Multi-Criteria Labeling Engine and a Prescriptive Advisory System, the pipeline successfully bridged the gap between raw data processing and practical, localized field deployment. The system translates abstract mathematical coordinates into actionable, real-world blueprints—such as prescribing lime applications for acidic highland soils or freshwater leaching protocols for saline Rift Valley lowlands.

8.2. Model Limitations

1. **Extreme Outlier Gravitation:** K-Means clustering relies fundamentally on minimizing squared Euclidean distances. As discovered during data diagnostics, severe regional soil hazards—such as the lowland salinity spikes reaching $\approx 16,000$ uS/cm—exert a massive mathematical pull on moving centroids. Consequently, the algorithm acts more like an anomaly quarantine mechanism.
2. **Rigidity of the Rule-Based Layer:** The prescriptive advisory engine relies on static, hardcoded thresholds to generate its remediation text. While these thresholds are aligned with agronomic standards, they lack the dynamic adaptability of a pure machine learning model.
3. **Assumption of Spherical Cluster Geometry:** K-Means implicitly assumes that data groups are isotropic and spherically distributed. Real-world soil telemetry frequently exhibits complex, non-linear, and elongated density shapes due to compounding environmental factors.

8.3. Future Extensions

1. **Transition to Density-Based Clustering (DBSCAN):** Future iterations will replace K-Means with DBSCAN. DBSCAN does not require a pre-configured cluster count and defines clusters based on spatial density. This switch will allow the model to natively isolate extreme anomalies as noise points, protecting primary cluster groups.
2. **Multi-Modal Telemetry Fusion:** Soil chemistry does not exist in a vacuum. The pipeline will be expanded to combine chemical data with localized spatial and environmental layers (Elevation, Slope, Mean Rainfall, Surface Temperature).
3. **Hybrid Supervised-Unsupervised Pipeline Integration:** The final evolution of this framework will connect this unsupervised pipeline with a supervised classifier (such as a Random Forest Classifier).

trained on crop-specific yields) to predict exact crop compatibility scores for regional staples like Teff, Wheat, Sorghum, and Maize.

9. References & Appendices

9.1. References

Ethiopian Soil Information System (EthioSIS). (2021). Soil Fertility Status and Fertilizer Recommendation Reports for the Ethiopian Highlands. Ministry of Agriculture, Addis Ababa.

MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, 1, 281-297.

Tadesse, G., & Shiferaw, B. (2023). Managing Soil Salinity and Alkalinity Hazards in the Awash River Basin Irrigation Schemes: Technological and Policy Options. *Ethiopian Journal of Agricultural Sciences*, 33(2), 45-62.

Assefa, S., & Spatial Soil Audit Team. (2024). Acidification trends in high-rainfall agro-ecologies of Western Ethiopia: Nutrient fixation mechanics and liming efficiency. *Geoderma Regional*, 36, e00712.

Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.